NAVAL, Julia Marie P.
BSIS02

Provide a clear definition and overview of the following:

Data Structures and Algorithms

Data structures are ways of organizing and storing data so a computer can access and use it efficiently. Common examples include arrays, stacks, queues, linked lists, trees, and graphs.

Algorithms are step-by-step instructions used to solve a problem or complete a task. Data structures and algorithms work together because the way data is organized affects how efficiently an algorithm can process it.

Linear Data Structure

A linear data structure stores elements in a sequential order. Each element is arranged one after another. Examples include arrays, linked lists, stacks, and queues.

One-Dimension Array

A one-dimensional array stores multiple values of the same type in a single sequence. Each value has an index, which usually starts at 0. The array has a fixed length when it is created in languages such as Java.

The Big O Notation

Big O notation describes how the time or memory required by an algorithm grows as the amount of input increases. It helps programmers compare the efficiency of algorithms.

Common examples include:

O(1), constant time. The operation takes roughly the same amount of work regardless of the input size.

O(n), linear time. The amount of work increases with the number of elements.

O(log n), logarithmic time. The amount of work grows slowly as the input increases. Binary search has O(log n) worst-case time when searching a sorted collection.

Basic Searching Algorithms

Searching algorithms are used to find a specific value or element within a collection of data. Two basic searching algorithms are linear search and binary search.

Linear Search

Linear search checks each element one at a time from the beginning until it finds the target or reaches the end of the collection. It can be used even when the data is not sorted.

Binary Search

Binary search works by repeatedly dividing a sorted collection into two parts. It checks the middle element and removes the half that cannot contain the target. The data must be sorted before using binary search.